RoadEnv: an agent evaluation environment over a real, messy government report

Pradeep Rathore · proof of work for Member of Technical Staff, Environment Engineering · built and run Oct 2026 · code: github.com/Pradeep00777/roadenv

30 tasks	**130 of 266**	**4 → 12 / 30**	**38%**
hand-verified answers over a 266-page MoRTH report	pages have no text layer, including all 50 annexure tables	naive agent, then the same agent handed the right page	of answerable tasks have a gold page in search top-3

PROBLEM

Labs and enterprises need agents that answer from real documents, and need to *measure* whether they do. Public reports are a hard, realistic case: India's *Road Accidents in India 2024* uses Indian digit grouping (4,87,707), splits tables across pages with no title on the continuation page, wraps row labels, marks missing cells `NA` with footnotes, and puts every state-level annexure in image-only pages (some rotated).

ENVIRONMENT DESIGN

- **Episode** = one question, 10 tool calls: `search` (BM25 over pages), `read_page`, `calculate`, `submit(answer, status, citation_page)`. Full trace logged.
- **Two corpora**: the PDF's own text layer, and text + tesseract OCR (auto-rotation) for the 102 image-only annexure and appendix pages.
- **Tasks**: 13 lookups, 4 layout traps, 1 unit, 4 derived, 3 convention, 2 image-only, 3 unanswerable (incl. a near-miss: per-victim compensation amounts exist, totals paid do not).
- **Plug-in LLM agent** for any OpenAI-compatible API is included; no model was run here (no API key in the build sandbox).

GRADER (DETERMINISTIC)

- Parses 4,87,707 / 4.71 lakh / –1.38%; exact match for counts, stated tolerance for rates.
- Refusing an answerable task and answering an unanswerable one are scored as separate failure types.
- A citation counts only if the cited page has a line matching the task's evidence regex. Reward = correct × (0.8 + 0.2 × valid citation). Self-test: gold answers score 30/30; known-wrong answers fail.

BASELINE RESULTS (REAL RUNS)

agent	corpus	correct	look	messy	deriv	spec	image	unans.
refuse-all	text	**4/30**	0/13	1/4	0/4	0/3	0/2	3/3
retrieve + regex	text	**4/30**	0/13	2/4	0/4	0/3	0/2	2/3
retrieve + regex	text+OCR	**5/30**	1/13	2/4	0/4	0/3	0/2	2/3
same, gold page	text	**12/30**	8/13	2/4	0/4	0/3	0/2	2/3

Search recall@10: 69% on the text layer, 54% after adding OCR. The baselines calibrate the env: refuse-all scoring 13% shows that refusing doesn't pay.

FAILURE MODES FOUND

1. **Retrieval, not reading, is the bottleneck**: 4/30 rises to 12/30 when the right page is given. State master tables name each state once, so page-level search prefers prose pages.
2. **Blending OCR into one index hurt**: recall@10 fell 69% → 54%; image-only tasks stayed 0/2.
3. **OCR corrupts cells silently**: Annexure 5 reads D&N Haveli's 2023 value 73 as "2B".
4. **The report has a printed error**: Goa's 2024 change in deaths is printed –1.28%; its own counts (290 → 286) give –1.38%.
5. **Conventions must be pinned**: "1,336 accidents a day" divides by 365, but 2024 had 366 days (1,333).
6. **Right row, wrong cell**: rank read as count, share as count, a paragraph number read as a value.
7. **Year words lure hallucination**: "accidents in 2025" was answered from a page citing *World Road Statistics 2025*.

WHAT I WOULD DO IN WEEK ONE AT A CUSTOMER

1. **Days 1–2:** sit with the people who do the task today; collect 30–50 real questions with the answers they accept, and write down their conventions (rounding, which table wins, what "per day" means) before writing a grader.
2. **Days 2–3:** build the environment on their real documents and tools; deterministic grader with an evidence rule; run trivial baselines first. If refuse-all scores well, the task set is broken.
3. **Days 3–4:** run their model; sort failures by root cause (retrieval, cell reading, units, conventions, image-only, hallucination) and fix the top two, in the data or in the environment.
4. **Day 5:** report the numbers, say plainly what is not measured yet, and agree on the next 100 tasks.

Source: MoRTH, Road Accidents in India 2024 (June 2026), morth.gov.in/backend/documents/uploaded/1781177676_V1gUW8tJWT.pdf. Limits: baselines only, no LLM run; 30 tasks, one document, single annotator; image-only answers checked visually on the rendered page. The 36 state/UT rows of Tables 5.1 and 5.6 do sum exactly to the All-India totals for 2020–2024.